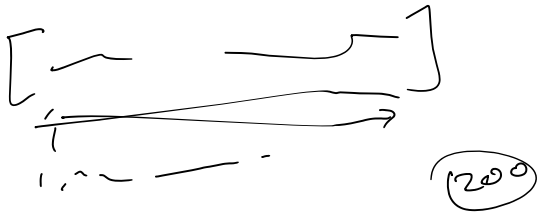


searching
 ↳ Linear ↳ Binary

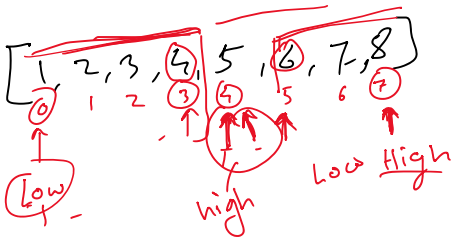


$$T = \tau \quad (100)$$

A B C D

Discovery

B A Q E D
 ↳ E



$$\begin{aligned} &\rightarrow \frac{0+7}{2} = 3 \\ &\frac{4+7}{2} = 5 \\ &\frac{4+4}{2} = 4 \end{aligned}$$

```

while (low <= high) {
    mid = (low + high) / 2
    if (arr[mid] == target) {
        return mid
    }
    if (arr[mid] > target) {
        high = mid - 1
    }
    else {
        low = mid + 1
    }
}
return -1

```

$$x^k < n$$

$$n = 149$$

$$\underline{x^k \leq n}$$

$$n = 149$$

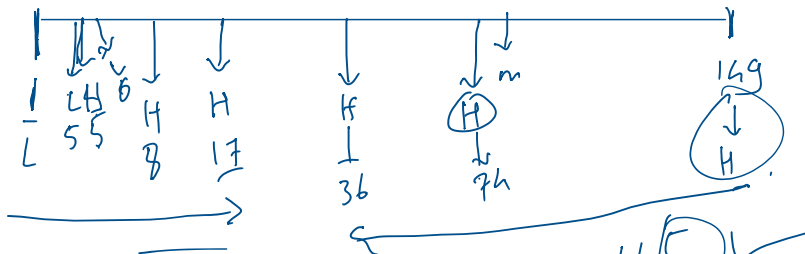
$$\underline{k = 3}$$

$$x = 1$$

```
while (xk ≤ n) {
    x++
}
```

cout (x-1)

$$\begin{aligned} &\rightarrow 1^3 \leq 149 \\ &\rightarrow 2^3 \leq 149 \\ &\rightarrow 3^3 \leq 149 \\ &\rightarrow 4^3 \leq 149 \\ &\rightarrow 5^3 \leq 149 \\ &\rightarrow 6^3 \leq 149 \end{aligned}$$



$$ans = 45$$

$$7^3 \leq 149$$

$$37^3 \leq 149$$

$$18^3 \leq 149$$

$$9^3 \leq 149$$

$$4^3 \leq 149$$

$$\begin{aligned} &\rightarrow 4^3 \leq 149 \\ &6^3 \leq 149 \\ &5^3 \leq 149 \end{aligned}$$

$$\frac{5+8}{2} = 6$$

$$\frac{5+5}{2} = 5$$

$$\frac{1+149}{2} = 75$$

$$\frac{1+75}{2} = 37$$

$$\frac{1+36}{2} = 18$$

$$\frac{1+17}{2} = 9$$

$$\frac{1+8}{2} = 4$$

```
if (midk ≤ n) {
    ans = mid
    Low = mid + 1
}
```

```
else {
    High = mid - 1
}
```

Math.pow(mid, k)

3